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2023-3R/2024 (MAT3103SE)

2024

MATHEMATICS

Paper : MAT 3103SE

(Programming in Python)

Full Marks : 35

Time : 1½ hours

***The figures in the margin indicate
full marks for the questions.***

1. Answer to the following : 1×4=4
- (a) What is a dictionary in Python, and how does it differ from a list ?
 - (b) Why is * called string repetition operator ?
 - (c) Write the purpose of **stamp** () method.
 - (d) What is the purpose of exit functions in Python ?

Contd.

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2. Answer to the following briefly :

2×3=6

- (a) Using the range function, write a loop that prints numbers from 0 to 57.
- (b) Given the dictionary student =
{ "name": "Alice", "age": 20, "grade": "A" } ,
write a command to update Alice's grade to "A+" and retrieve her age.
- (c) Explain the difference between **read** () and **readlines** () methods when reading a file in Python. In what situations might you prefer one over the other ?

3. Answer **any three** of the following :

5×3=15

- (a) What is the difference between the else block and the finally block in exception handling ? Explain with an example program.
- (b) Write a program using Python as a calculator.
- (c) Write a simple Python program using a **for loop** to print all even numbers from 1 to 20. Explain how the loop works in your code.

(d) Write a brief note on PIP. Explain installing packages via PIP.

(e) Write a Python program to convert height in feet and inches to cm.

[1feet = 12inch and 1inch = 2.54cm]

(Sample input : 2 feet 7 inch, Sample output : 78.74 cm)

4. Answer **any one** of the following :

10×1=10

(a) Write a Python program that prompts the user to enter a series of numerical values (e.g., monthly temperature variation) and then plots a line graph based on the input. Label the axes add a title to the graph. Then, customize the graph by changing the line style, color, and adding data point markers.

Or

(b) Define and describe the role of algorithms and flowcharts in problem-solving. Create a flowchart and algorithm for a Python program that takes an integer input from the user and checks whether it is a prime number.
